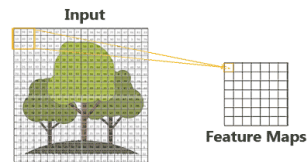




Convolutional Neural Networks (Transfer Learning)

If you already know how to drive a car, how hard is it to learn to drive a truck?



Dr. Anish Monsley K.,
PostDoctoral Researcher, IIT Madras

David Perkins

1992.....Just a history

- Educational psychologist, Harvard Graduate School of Education
- Founding member of **Project Zero**
- Research focus: **How knowledge transfers across tasks**

Key Contribution

- Introduced the concept of **Transfer of Learning**
- Showed that transfer is **not automatic** and depends on how knowledge is learned

Two Types of Transfer (Perkins)

- **Low-Road Transfer**: Automatic transfer based on similarity
→ Corresponds to **feature extraction in transfer learning**
- **High-Road Transfer**: Deliberate abstraction and adaptation
→ Corresponds to **fine-tuning and model adaptation**

Relevance to AI Transfer Learning

- Prior human knowledge $\leftrightarrow$ **Pretrained model weights**
- Failure of transfer $\leftrightarrow$ **Negative transfer**
- Abstraction of ideas $\leftrightarrow$ **Representation learning**



He is **not a machine-learning researcher**, but his ideas are **directly relevant to Transfer Learning (TL)** at a *conceptual and pedagogical level*—which is why he is often cited in learning sciences, education, and AI-education discussions.

Prof. Fei-Fei Li

ImageNet was developed by Prof. Fei-Fei Li and her research team at Princeton University, later continued at Stanford University.

Key people involved

- **Prof. Fei-Fei Li** – Lead creator and principal investigator
- **Kai Li**
- **Jia Deng**
- **Wei Dong**
- **Richard Socher** (later contributions)



Timeline

- **2007** – ImageNet project initiated
- **2009** – ImageNet dataset released
- **2010–2017** – ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

ImageNet Dataset

What is ImageNet?

- Large-scale visual database created for image recognition research
- Contains **14+ million labeled images** across **20,000+ object categories**
- Organized using **WordNet hierarchy**

Why ImageNet Was Revolutionary

- Enabled training of **very deep CNNs**
- Provided sufficient data to learn **general visual features**
- Led to breakthroughs such as **AlexNet (2012)**

Impact on Deep Learning

- Standardized benchmark for vision
- Foundation for most pretrained CNNs (VGG, ResNet, EfficientNet)
- Made transfer learning **practical, scalable, and reliable**

How ImageNet Enabled Transfer Learning

- Models trained on ImageNet learn:
 - Edges, textures, shapes (early layers)
 - Object-level semantics (deep layers)
- These learned representations can be **reused across tasks**

Limitations of Existing Techniques: Why Transfer Learning Emerged

Before Transfer Learning

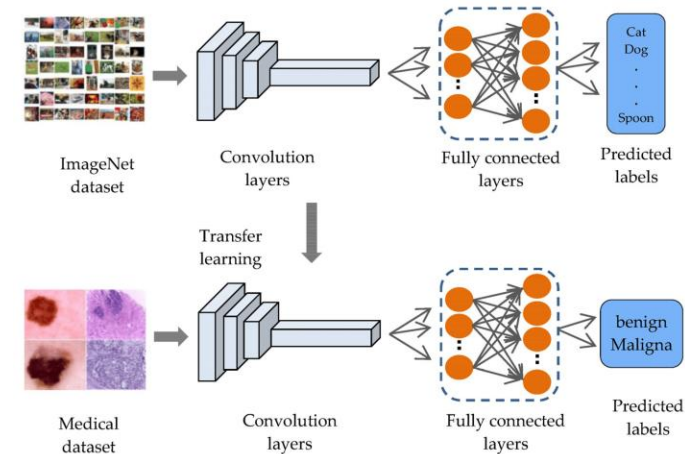
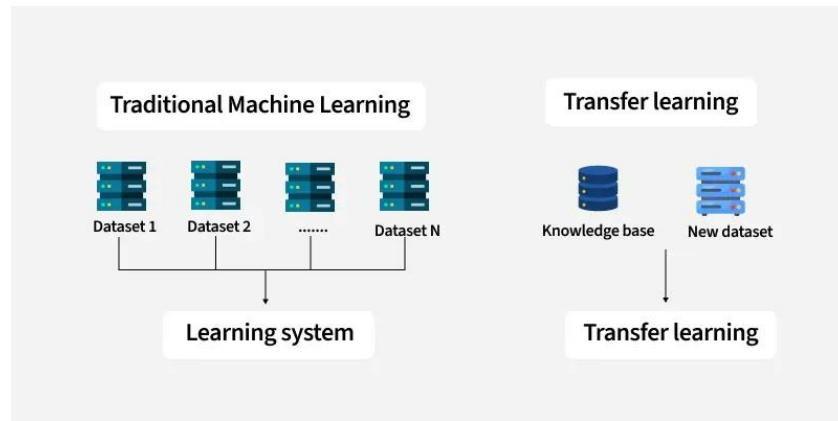
- Models were trained **from scratch** for every new task
- Required **large labeled datasets** for each problem
- Training deep models was **time-consuming and expensive**
- Learned knowledge was **not reusable**
- Models often **overfitted** when data was limited

Key Problems

- Data scarcity in many real-world domains (medical, industrial, logistics)
- High computational cost
- Poor generalization to new but related tasks

Introduction to Transfer Learning

- Deep learning models require large datasets
- Training from scratch is expensive and slow
- Humans reuse knowledge across tasks
- **Transfer Learning** enables models to reuse learned knowledge



What is Transfer Learning?

Transfer Learning is the process of using a model trained on one task (source task) and adapting it to a different but related task (target task).

Formal view

- Source domain: $D_s=(X_s,Y_s)$
- Target domain: $D_t=(X_t,Y_t)$
- Goal: Improve learning of $f_t(\cdot)$ using knowledge from $f_s(\cdot)$

Same domain, different task (ImageNet $\rightarrow$ medical images)

Different domain, same task (natural images $\rightarrow$ satellite images)

Where Transfer Learning is Used?

Domain	Example
Computer Vision	ImageNet → Face Recognition
NLP	BERT → Sentiment Analysis
Speech	Pretrained ASR → Call-center audio
Medical	Natural images → X-rays
Logistics	Traffic images → Warehouse safety

CNN Perspective: *What exactly is transferred?*

CNN Layer	Learns
Early layers	Edges, corners
Middle layers	Textures, shapes
Deep layers	Objects, semantics

Early layers are **generic**, deeper layers are **task-specific**

Types of Transfer Learning

Type 1: Feature Extraction (Freezing)

- Freeze pretrained layers
- Train only classifier head
- Best when:
 - Small dataset
 - Similar domain

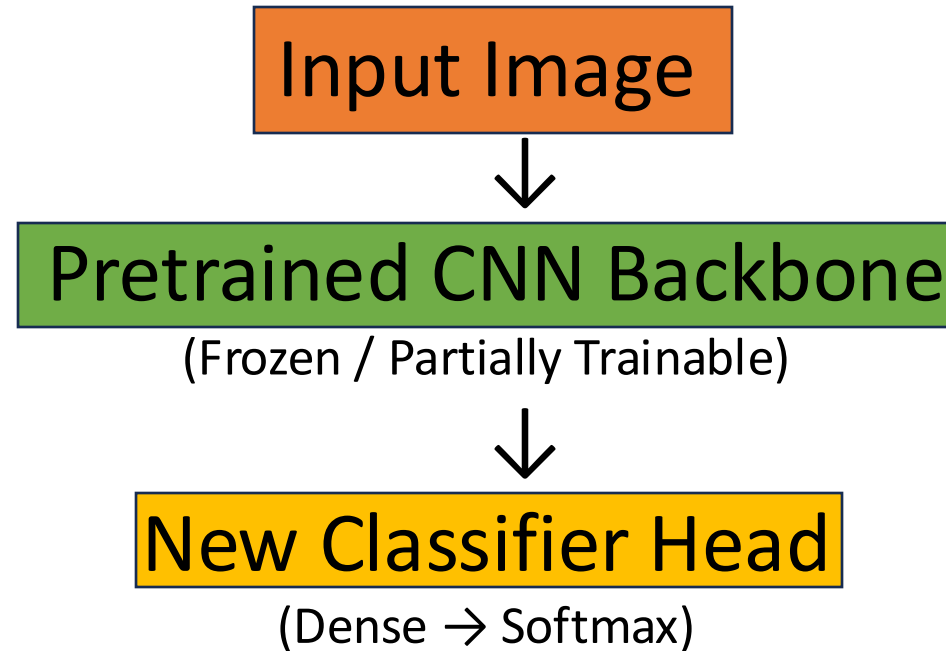
Type 2: Fine-tuning

- Unfreeze some layers
- Train with low learning rate
- Best when:
 - Larger dataset
 - Target task slightly different

Type 3: Full Training

- Rarely recommended
- Needs large dataset

Architecture



Thus, we keep the brain !, replace the decision-maker

Type 1: Feature Extraction (Freezing)

Feature-Based Transfer Learning == Using Pretrained Model as a Fixed Feature Extractor

- A pretrained model already knows **how to extract useful features** from data
- We simply **reuse those features** and train a new decision layer.

→ What happens internally?

- Convolution layers act as **feature extractors**
- These layers are **not updated**
- Only a small network at the end learns the new task

→ Why does this work?

Deep models learn **general patterns first**:

- Edges
- Shapes
- Textures

These patterns are **common across many problems**.

When is this the best choice?

- You have **limited training data**
- Your task is **similar** to what the model originally learned
- You want **fast and stable results**

Example

- Model trained on ImageNet
- New task: Detect helmets in warehouse images
- The model already understands:
 - Human shapes
 - Objects
 - Background separation

Type 1 ---->>>>

“You are not teaching someone how to think — only how to decide.”

Type 2: Fine-tuning

Fine-Tuned Transfer Learning == Partial Adaptation

- Some of the pretrained knowledge is useful, but **not perfectly aligned** with the new task.
- We allow the model to **slightly adjust itself**.

What changes?

- Lower layers remain unchanged
- Higher layers are updated
- Learning happens **slowly and carefully**

Why only higher layers?

- Early layers detect universal patterns
- Deeper layers capture task-specific semantics
- Modifying only deeper layers prevents damage to general knowledge

Why low learning rate?

- Pretrained weights are already close to optimal.
- Large updates would **erase useful information**.

“General Features → Frozen”
“Task-Specific Features → Trainable”

When should you fine-tune?

- You have **moderate amount of data**
- Your task differs slightly from the original one
- Accuracy improvement is important

Example

- Natural images → Medical scans
- Road vehicles → Warehouse forklifts

Type 2 ---->>>>

“You keep the foundation, but renovate the upper floors.”

Type 3: Full Training

Full Model Retraining == Complete Adaptation

- The pretrained model provides a **starting point**, but **everything is allowed to change**.

What happens?

- Entire network is trainable
- Model fully adapts to new data
- Training resembles building a new model, but faster convergence

Why is this uncommon?

- Deep networks have millions of parameters
- Requires enormous data
- High risk of overfitting

When does this make sense?

- Very large dataset available
- New task is very different from original
- Strong computational resources available

Example

ImageNet → Satellite imagery
RGB photos → X-ray scans

Type 3---->>>>

“You unlearn and relearn everything — with some prior experience.”

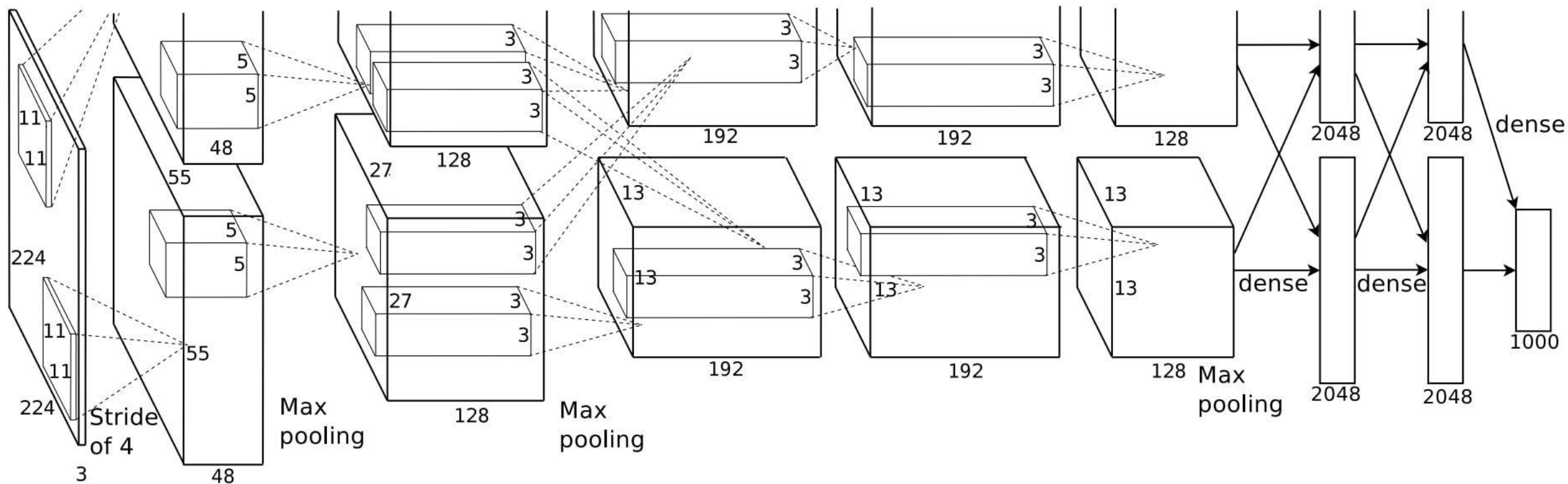
Just a comparison table...

Approach	Knowledge Reuse	Data Needed	Flexibility
Feature-Based	High	Low	Low
Fine-Tuned	Medium	Medium	Medium
Full Retraining	Low	Very High	High

The more data you have, the more freedom you can give the model.

I have added a list of popular pre-trained models for your reference...

AlexNet



Popular Pretrained Models

Model Family	Models Included	Key Idea	Typical Use
Classic CNNs	LeNet, AlexNet	Early deep CNNs	Learning basics
VGG	VGG16, VGG19	Deep + uniform conv layers	Transfer learning
Residual Networks	ResNet-18/34/50/101/152	Skip connections	General-purpose
Dense Networks	DenseNet-121/169/201	Feature reuse	Efficient learning
Lightweight Models	MobileNet v1/v2/v3, SqueezeNet, ShuffleNet	Low compute	Mobile & edge
Efficient Models	EfficientNet B0–B7, EfficientNet-V2	Compound scaling	High accuracy
Modern CNNs	ConvNeXt	CNN + transformer ideas	State-of-the-art

Vision Transformers & Hybrid Models

Model Family	Models Included	Core Idea	Strength
Vision Transformer	ViT-B/L/H	Pure transformer	Global context
Data-efficient ViT	DeiT	Efficient training	Better transfer
Hierarchical ViT	Swin Transformer	Window-based attention	Detection & segmentation
Self-supervised ViT	BEiT	Masked modeling	Strong representations
Hybrid Models	CvT	CNN + Transformer	Efficient vision tasks

Object Detection – Pretrained Models

Detection Type	Models	Key Property	Typical Use
Two-stage	Faster R-CNN, Mask R-CNN, Cascade R-CNN	High accuracy	Research & medical
One-stage	YOLO v3–v8, SSD, RetinaNet	Real-time	Surveillance, robotics
Keypoint-based	CenterNet	Anchor-free	Lightweight detection

Image Segmentation – Pretrained Models

Segmentation Type	Models	Strength	Domain
Medical Segmentation	U-Net	Works with small data	Healthcare
Semantic Segmentation	DeepLab v3/v3+, PSPNet	Scene understanding	Autonomous driving
High-resolution	HRNet	Precise boundaries	Fine-grained tasks
Transformer-based	SegFormer	Efficient + accurate	Modern vision

Face Recognition – Pretrained Models

Model	Core Idea	Common Use
VGG-Face	CNN-based face features	Face verification
FaceNet	Embedding learning	Face matching
ArcFace	Margin-based loss	High-accuracy systems
DeepFace	Deep identity learning	Large-scale systems
InsightFace	Optimized face embeddings	Production use

Self-Supervised Vision Models

Model	Pretraining Method	Advantage	Transfer Learning Use
SimCLR	Contrastive learning	Simple & strong	Small labeled data
MoCo / MoCo-v3	Momentum contrast	Stable training	General vision
BYOL	No negative samples	Robust features	Vision transfer
DINO	Self-distillation (ViT)	Excellent features	Detection & segmentation
MAE	Masked autoencoding	Efficient pretraining	Large-scale vision

Natural Language Processing – Pretrained Models

Model Type	Models	Main Use
Encoder	BERT, RoBERTa, ALBERT, DistilBERT, ELECTRA	Classification
Decoder	GPT-2/3/4, OPT	Text generation
Encoder–Decoder	T5, BART, FLAN-T5	Translation, summarization

Speech & Audio – Pretrained Models

Model	Task	Key Feature
Wav2Vec 2.0	Speech recognition	Self-supervised
HuBERT	Speech representation	Robust features
DeepSpeech	ASR	End-to-end
YAMNet	Audio classification	Lightweight

Multimodal (Vision + Language) Models

Model	Modalities	Typical Use
CLIP	Image + Text	Zero-shot learning
BLIP / BLIP-2	Image + Text	Captioning, VQA
ALIGN	Image + Text	Contrastive learning
Flamingo	Vision + Language	Few-shot tasks

Just adding few information to give a complete overview of Transfer learning

Common Pretraining Datasets

Dataset	Domain	Used For
ImageNet	Images	CNNs, ViTs
COCO	Detection	Object detection
OpenImages	Images	Large-scale vision
LibriSpeech	Audio	Speech models
Wikipedia + Books	Text	NLP models

Some notable reads

1. **Pan, S. J., & Yang, Q.** (2010). *A Survey on Transfer Learning* — IEEE Transactions on Knowledge and Data Engineering (highly cited foundational survey).
2. **Weiss, K., Khoshgoftaar, T. M., & Wang, D.** *A survey of transfer learning* — Journal of Big Data (comprehensive survey of methods & types).
3. **Taylor, M. E., & Stone, P.** (2009). *Transfer Learning for Reinforcement Learning Domains: A Survey* — Journal of Machine Learning Research.
4. **Ruder, S., Peters, M. E., Swayamdipta, S., & Wolf, T.** (2019). *Transfer Learning in Natural Language Processing* — NAACL overview of modern NLP transfer learning models and methods.
5. **Zhuang, F., et al.** (2021). *A Comprehensive Survey on Transfer Learning* — Proceedings of the IEEE (recent large survey covering many modern TL approaches).
6. **Torrey, L., & Shavlik, J.** (2010). *Transfer learning*, in *Handbook of research on machine learning applications and trends: algorithms, methods, and techniques*. (classic book chapter).

Some useful links

- <https://github.com/axruff/TransferLearning>
- <https://github.com/thuml/Transfer-Learning-Library>

Thank You

All the contents of this presentation have been prepared based on lecture slides by **Prof. Babji Srinivasan, IIT Madras**, along with information drawn from **open-source materials** available in the public domain.